

NovaMart E-Commerce Sales Data Analysis Project

Client Problem Statement & Business Requirements

1. Client Information

Client Name:	NovaMart
Industry:	E-Commerce / Online Retail
Project Type:	Sales Data Analysis using SQL
Your Role:	Data Analyst
Data Period:	January 2023 – June 2025

2. Business Background

NovaMart is a growing online multi-category retailer that sells products across five major categories:

- **Electronics** — Smartphones, Laptops, Headphones, Smartwatches, etc.
- **Clothing** — Men, Women & Kids Apparel, Footwear, Accessories
- **Home & Kitchen** — Cookware, Appliances, Furniture, Decor
- **Sports & Outdoors** — Fitness Equipment, Outdoor Gear, Cycling, Camping
- **Beauty & Personal Care** — Skincare, Haircare, Makeup, Fragrances

The company has been collecting sales data for the past 2.5 years. Management now wants a clear understanding of business performance so they can make better data-driven decisions regarding inventory, marketing, pricing, and customer retention.

3. Data Provided by Client

The client has shared three related tables:

Table	Records	Description
Customers	500	Customer details: Name, Email, City, Country, Gender, Age, Signup Date
Products	199	Product catalog: Name, Category, Brand, Unit Price, Cost Price, Stock
Orders	850	Sales transactions: Customer, Product, Date, Quantity, Amount, Status, Payment Method

4. Project Objective

As a Data Analyst, your task is to analyze the sales data using SQL and provide clear, actionable insights that will help NovaMart management understand:

- Overall sales performance and revenue trends
- Best and worst performing products, brands, and categories
- Customer purchasing behavior and high-value customers
- Product profitability (Gross Profit & Margin)
- Inventory health and stock-out risks
- Order patterns, cancellations, and payment method performance

Expected Output: Write SQL queries to answer all business questions listed below and summarize the key findings so management can take better decisions.

5. Business Questions to Solve

Below are the complete list of questions the client wants you to answer using SQL:

BASIC LEVEL

1. Retrieve all products belonging to the "Electronics" category.
2. Find all products with a Unit_Price greater than \$500.
3. List all customers from the United States.
4. Show all orders placed in November 2024.
5. Calculate the total stock available across all products.
6. Find the details of the most expensive product.
7. Show all orders where the customer ordered more than 3 quantity of a product.
8. Retrieve all orders where the Total_Amount exceeds \$200.
9. List all distinct categories and brands available in the Products table.
10. Find the product with the lowest stock.
11. Calculate the total revenue generated from all Completed and Shipped orders.

INTERMEDIATE LEVEL

12. Find the total quantity sold and total revenue for each product category.
13. Calculate the average price of products in the "Clothing" category.
14. List customers who have placed at least 3 orders.
15. Find the most frequently ordered product (by total quantity sold).
16. Show the Top 5 most expensive products in the Electronics category.
17. Calculate the total quantity sold by each Brand.
18. List the cities of customers who have spent more than \$300 in total.
19. Find the customer who spent the most (highest lifetime value).
20. Calculate the remaining stock of each product after fulfilling all Completed/Shipped orders.

ADVANCED LEVEL

21. Show the monthly revenue trend (Year-Month) for Completed and Shipped orders.
22. Calculate the revenue contribution percentage of each Category.
23. Calculate Gross Profit and Gross Margin % for each Category. (Profit = Total_Amount – Cost_Price × Quantity)
24. Find the Top 10 customers by total spend.
25. Calculate the cancellation rate by month.
26. Find the best performing brand in each category (using ranking).
27. Segment customers into High Value / Medium Value / Low Value based on total spending.
28. Identify products that are at stock-out risk (high sales quantity + low current stock).
29. Analyze performance by Payment Method (orders, average order value, and total revenue).
30. Calculate Month-over-Month revenue growth percentage.

Important Note for Students

Treat this as a real client project. First understand the business problem, then write clean SQL queries to answer each question, and finally summarize the key insights in simple language that non-technical managers can understand. This approach will make your project strong for both academic evaluation and your professional portfolio (Resume / GitHub / LinkedIn).

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